

Five stocks against each other — OLS, logistic regression and the limits of both

Five years of live daily data on Apple, Coca-Cola, JPMorgan, ExxonMobil and NVIDIA, analysed pairwise and against the S&P 500. Every estimate here was produced by this project's own OLS and logistic implementations, with heteroskedasticity-robust standard errors and strictly chronological validation. The headline result is not a ranking — it is that the *same method* generalises for three of these five names and fails completely for the other two, and the market model tells you in advance which is which.

THE UNIVERSE

AAPL 5-YEAR TOTAL +119% β 1.19 · vol 28%	KO 5-YEAR TOTAL +78% β 0.26 · vol 17%	JPM 5-YEAR TOTAL +155% β 0.89 · vol 24%	XOM 5-YEAR TOTAL +221% β 0.41 · vol 27%	NVDA 5-YEAR TOTAL +995% β 2.17 · vol 52%
---	--	--	--	---

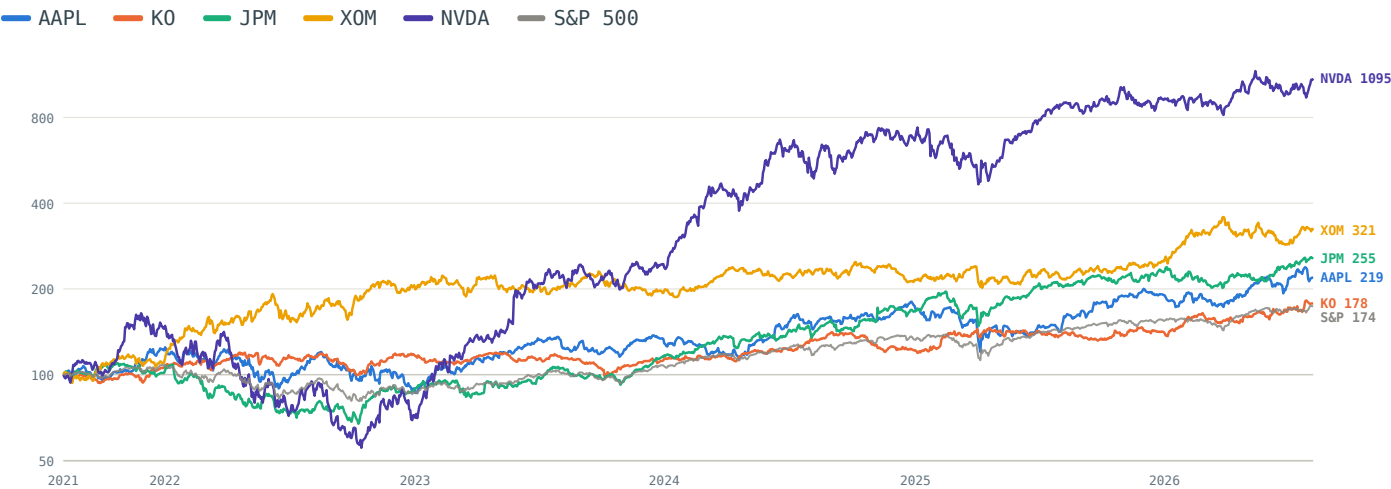
METHOD

All models use **daily simple returns**, never price levels — regressing one price series on another is a spurious regression, because both are non-stationary. Standard errors are **HC3 robust** throughout: a Breusch-Pagan test rejects homoskedasticity for every name in this set, so classical errors would understate the true uncertainty. Out-of-sample tests use a **chronological 70/30 split** with no shuffling, plus five-fold **expanding-window walk-forward** validation. Sample: 1,254 aligned trading days, 2021-08-10 to 2026-08-07, adjusted closes from Yahoo Finance.

GROWTH OF 100, LOG SCALE

All six series rebased to 100 at the start of the window

A logarithmic axis is required here, not cosmetic: NVIDIA ends near 1,100 while Coca-Cola ends near 178. On a linear axis the other five would be flat against the bottom. On a log axis equal vertical distances are equal *percentage* moves, so the slopes are directly comparable.



For each stock: $r_{i,t} = \alpha + \beta \cdot r_{mkt,t} + \varepsilon_t$. Beta measures sensitivity to the market; alpha is the return unexplained by it; R^2 is the share of daily variance the market accounts for. The residual share is idiosyncratic — company-specific risk that diversification can reduce.

SYMBOL	COMPANY	B	SE (HC3)	P (B)	R ²	IDIO.	A ANN.	P (A)	ANN. VOL	5Y TOTAL
AAPL	Apple Inc.	+1.192	0.044	5e-126	0.524	48%	+4.6%	0.592	28.1%	+119%
KO	Coca-Cola Company (The)	+0.263	0.039	2e-11	0.073	93%	+9.7%	0.187	16.6%	+78%
JPM	JP Morgan Chase & Co.	+0.887	0.037	5e-104	0.383	62%	+10.6%	0.218	24.4%	+155%
XOM	ExxonMobil Holdings Corporation	+0.411	0.055	1e-13	0.069	93%	+21.8%	0.060	26.7%	+221%
NVDA	NVIDIA Corporation	+2.171	0.064	2e-179	0.508	49%	+34.1%	0.037	51.9%	+995%

READING THE BETAS

An 8× spread in market sensitivity across one small portfolio

NVIDIA's β of **2.171** and Coca-Cola's **0.263** differ by a factor of more than eight. A 1% move in the S&P is associated with a 2.2% NVIDIA move and a quarter-percent Coca-Cola move. That is the difference between a high-beta cyclical and a defensive staple, and it shows up again in annualised volatility — 51.9% versus 16.6%.

Only one alpha is nominally significant, and it does not survive correction

NVIDIA's alpha (+34.1%/yr, $p = 0.037$) is the only one below the conventional 0.05 threshold. But five hypotheses were tested here. Under a Bonferroni correction the threshold becomes $0.05 / 5 = \mathbf{0.010}$, and $p = 0.037$ does not clear it. ExxonMobil ($p = 0.060$) and the rest are not close. **This report finds no statistically defensible evidence of alpha in any of the five names.** Reporting the raw p-value alone would have implied otherwise — which is exactly how multiple-comparison errors enter published results.

Low R^2 is not a failed model — it is the finding

Coca-Cola (R^2 0.073) and ExxonMobil (0.069) are barely explained by the market at all; roughly 93% of their daily variance is idiosyncratic. Apple (0.524) and NVIDIA (0.508) are about half market-driven. Page 5 shows that this single number predicts which logistic classifiers survive out-of-sample testing and which collapse.

Correlation of daily returns

Diverging scale: blue for positive co-movement, red for negative, neutral grey at zero. Values are printed in every cell, so the matrix is readable without relying on colour.

	AAPL	KO	JPM	XOM	NVDA
AAPL	1	+0.254	+0.362	+0.174	+0.479
KO	+0.254	1	+0.181	+0.160	-0.020
JPM	+0.362	+0.181	1	+0.280	+0.327
XOM	+0.174	+0.160	+0.280	1	+0.065
NVDA	+0.479	-0.020	+0.327	+0.065	1

PAIRWISE REGRESSION SLOPES

β from regressing the row's returns on the column's

Unlike correlation, this matrix is *asymmetric*: regressing NVIDIA on Apple gives 0.886, while Apple on NVIDIA gives 0.259. Slope carries the ratio of volatilities; correlation does not.

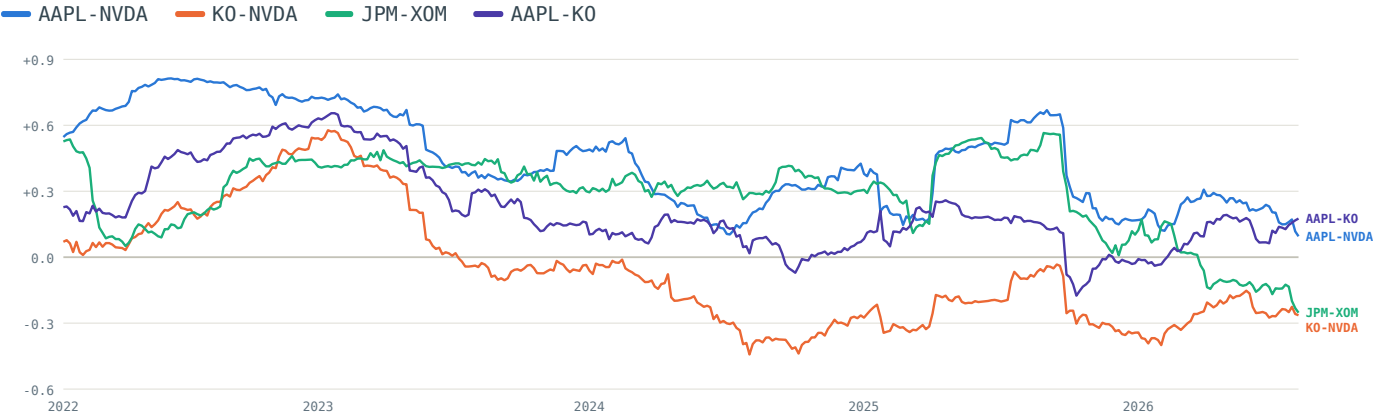
	AAPL	KO	JPM	XOM	NVDA
AAPL	—	+0.428	+0.415	+0.183	+0.259
KO	+0.150	—	+0.123	+0.100	-0.006
JPM	+0.315	+0.266	—	+0.257	+0.154
XOM	+0.165	+0.256	+0.305	—	+0.033
NVDA	+0.886	-0.063	+0.695	+0.126	—

The most defensive and the most aggressive name are uncorrelated

Coca-Cola against NVIDIA measures **-0.020** — statistically indistinguishable from zero over five years. ExxonMobil against NVIDIA is **+0.065**. These are the diversifying pairs in the set. The tightest pair is Apple–NVIDIA at **+0.479**, followed by Apple–JPMorgan at +0.362 — but page 4 shows those numbers are averages over regimes that look nothing alike.

120-day rolling correlation, four selected pairs

Each single number in the matrix on page 3 is the average of a series that swings enormously. Anything built on a static correlation assumption is built on a number that was true on average and rarely true at any given moment.



PAIR	FULL-SAMPLE CORR.	ROLLING MIN	ROLLING MAX	RANGE	LATEST
AAPL ↔ NVDA	+0.479	+0.095	+0.813	0.719	+0.095
KO ↔ NVDA	-0.020	-0.442	+0.577	1.019	-0.264
JPM ↔ XOM	+0.280	-0.253	+0.565	0.817	-0.253
AAPL ↔ KO	+0.254	-0.175	+0.655	0.830	+0.176

Apple-NVIDIA ranged from +0.81 to +0.10 — the same "0.48" the matrix reports

At its peak the pair moved almost in lockstep; at the latest reading it is close to independent. Coca-Cola–NVIDIA swung from +0.58 to −0.45, crossing zero repeatedly. A portfolio sized on the full-sample figure would have been badly mis-hedged in both directions at different times. This is the same instability that, in the market model, turns a strong in-sample relationship into an out-of-sample failure.

For each name: $P(\text{stock closes up}) = \sigma(\alpha + \beta \cdot r_{\text{mkt}})$, fitted by IRLS on the binomial log-loss — never squared error. Every model converged. The columns that matter are the last three: in-sample AUC is easy, out-of-sample AUC is the real test, and the majority-class baseline is what any model must beat to be worth anything.

SYMBOL	ODDS RATIO /+1%	P	MCFADDEN R ²	AUC IN-SAMPLE	AUC OUT-OF-SAMPLE	WALK-FWD MEAN	VERDICT
AAPL	5.38×	2e-54	0.250	0.842	0.761	0.788	holds up
KO	1.55×	6e-13	0.034	0.685	0.437	0.588	collapses
JPM	3.67×	5e-46	0.184	0.787	0.736	0.759	holds up
XOM	1.48×	4e-11	0.028	0.635	0.474	0.576	collapses
NVDA	6.91×	5e-58	0.290	0.843	0.815	0.819	holds up

The market-model R² predicts which classifiers survive

Rank the five by market R²: NVIDIA 0.508, Apple 0.524, JPMorgan 0.383 — then a gap — then Coca-Cola 0.073 and ExxonMobil 0.069. The first three hold their out-of-sample AUC (**0.82, 0.76, 0.74**). The last two collapse to **0.44 and 0.47** — at or below a coin flip. The relationship the classifier learned for Coca-Cola and ExxonMobil was never stable enough to project forward, and the R² said so before any classifier was fitted.

Beating AUC 0.5 is still not beating the baseline

AUC measures ranking. Accuracy against the majority-class baseline measures usefulness, and they disagree. More importantly, none of this is a strategy: the market return used as the predictor is *contemporaneous*, so it is not knowable before the stock's own close. These models describe co-movement. They do not forecast.

Not a strategy. No transaction costs, spreads, slippage, borrowing costs, taxes or capacity constraints are modelled anywhere in this document. **Not causal.** Co-movement between securities is an association, not a mechanism. **Not forward-looking.** Every relationship shown here is measured on 2021–2026 data, and pages 2 and 4 demonstrate directly that these relationships change. **Not a complete universe.** Five names chosen for sector contrast are not a market, and the set was not selected by any pre-registered rule. **Survivorship.** All five companies existed and were listed throughout the window, which biases any performance reading upward.

Educational and research use only. Market data may be delayed and depends on third-party provider coverage. This is not investment advice. No brokerage integration, order placement or automated trading is provided.